PROJECT SCHEDULE & RESOURCE PLANNING

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
Team Members	G Uday Kumar, Kavali Anu, Pavan Kumar Sowdepalli, Dasarayyagari Sravan Kumar
Maximum Marks	4 Marks

The following schedule outlines the planned development activities, estimated duration, project milestones, resource allocation, and potential project risks associated with the **OptiCrop: Smart Agricultural Production Optimization Engine**. The plan ensures systematic execution, efficient resource utilization, timely completion, and successful deployment of the proposed solution.

1. Project Timeline

#	Activity	Estimated Duration	Deliverable
1	Requirement Analysis	1 Week	Software Requirement Specification (SRS)
2	Dataset Collection and Analysis	1 Week	Prepared and Verified Agricultural Dataset
3	Data Pre-processing and Feature Engineering	1 Week	Cleaned and Processed Environmental Soil Matrix
4	Machine Learning Model Development	2 Weeks	Trained and Evaluated Crop Classification Models
5	Flask Web Application Development	2 Weeks	Functional Web Optimization Application
6	System Testing and Validation	1 Week	Testing and Validation Report
7	Deployment and Documentation	1 Week	Deployed Engine and Final Project Documentation

2. Major Milestones

Milestone	Expected Outcome
Requirement Analysis Completed	Approved Project Requirements Specification baseline.
Dataset Prepared	Training-ready cleaned biochemical (N, P, K, pH) and climate dataset.
Machine Learning Completed	Best performing predictive optimization model selected.
Web Application Developed	Crop recommendation interface successfully integrated with Flask backend.
System Testing Completed	Verified functional and reliable system performance report.

Milestone	Expected Outcome
Deployment Completed	Application successfully hosted on cloud infrastructure and ready for demonstration.

3. Risk Assessment & Mitigation

Identified Risk	Mitigation Strategy
Poor Data Quality	Perform thorough pre-processing, range validation, and strict outlier cleaning on inputs.
Low Prediction Accuracy	Train multiple machine learning algorithms (Random Forest, XGBoost) and select the best ensemble configuration.
Deployment Failure	Test cloud deployment thoroughly using Docker containers before moving to production clusters.
Security Vulnerabilities	Implement secure parameter validation, role-based database permissions, and encrypted token communications.
Project Schedule Delays	Monitor milestones regularly and distribute tasks efficiently across the five development team members.

4. Resource Allocation

Resource Layer	Technology Used	Purpose
Programming Core	Python 3.x	Machine learning model development, validation pipeline logic, and backend programming.
Backend Framework	Flask	Manages REST API endpoints, business logic processing, routing, and optimization services.
Frontend UI	HTML5, CSS3, JavaScript, Bootstrap 5	Builds a responsive, user-friendly field dashboard for checking crop suitability scores.
Database Stack	MySQL	Stores farmer accounts, regional land characteristics, and historical optimization logs securely.
ML & Science Engines	Scikit-learn, Pandas, NumPy	Supports comprehensive soil data parsing, scaling matrices, training cycles, and evaluation scores.
Development Tools	VS Code, Jupyter Notebook, Joblib, Postman	Provides system development workspace, pipeline testing, serialization, and endpoint API validations.
Version Control	Git and GitHub	Manages team source code, tracking changes, documentation versioning, and collaborative syncs.
Cloud Containerization	Docker & Cloud Services	Packages applications into robust portable micro-containers for repeatable cross-platform execution.

Conclusion

The project planning strategy provides a structured roadmap for successfully implementing the **OptiCrop: Smart Agricultural Production Optimization Engine**[cite: 14]. By precisely defining project phases, major milestones, resource allocation limits, cloud deployment strategies, and proactive risk management practices[cite: 14], the plan guarantees efficient technical development, exceptional prediction performance accuracy, adherence to timelines, secure data handling paradigms, and dependable remote environment cloud execution[cite: 14].